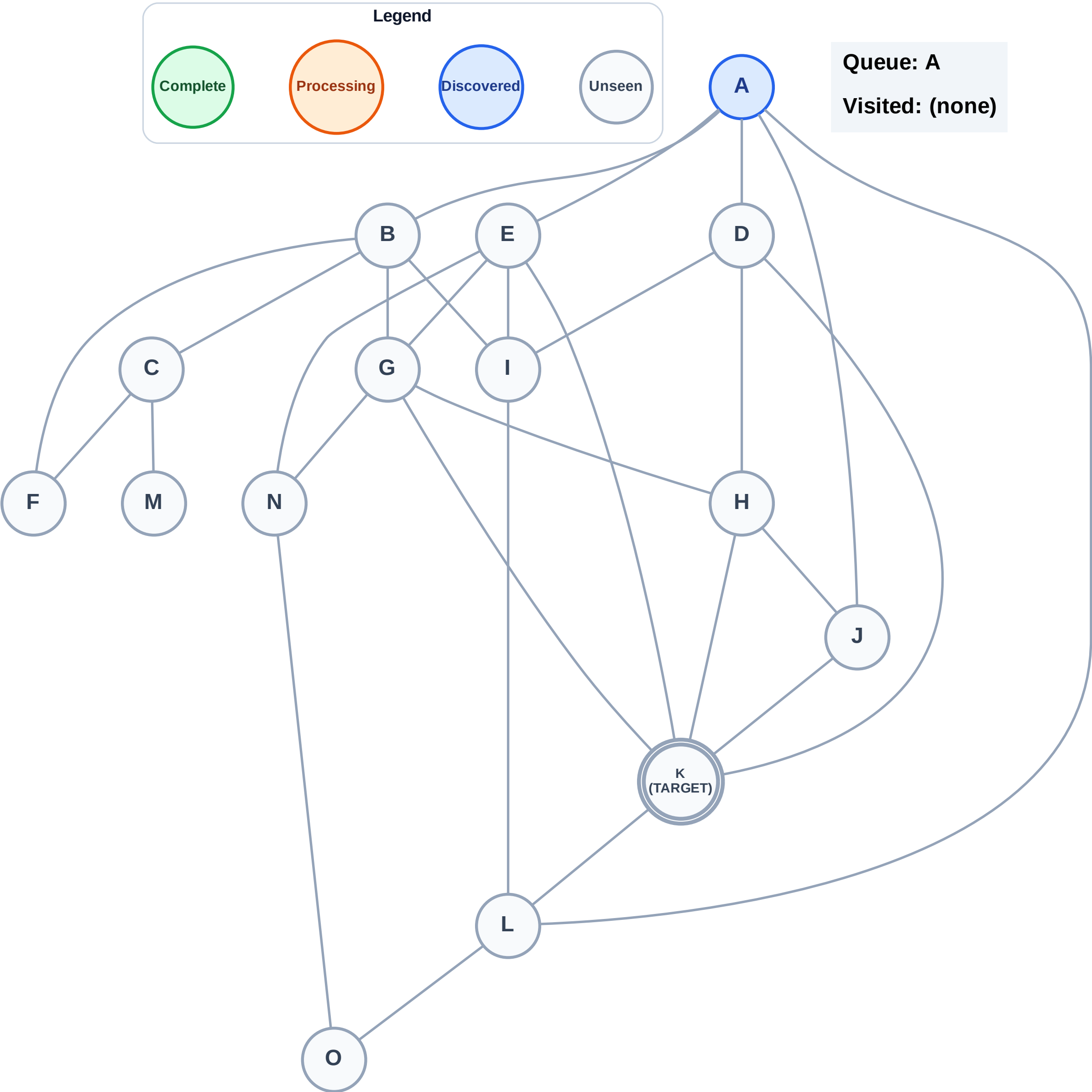


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



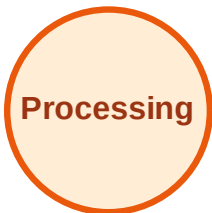
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



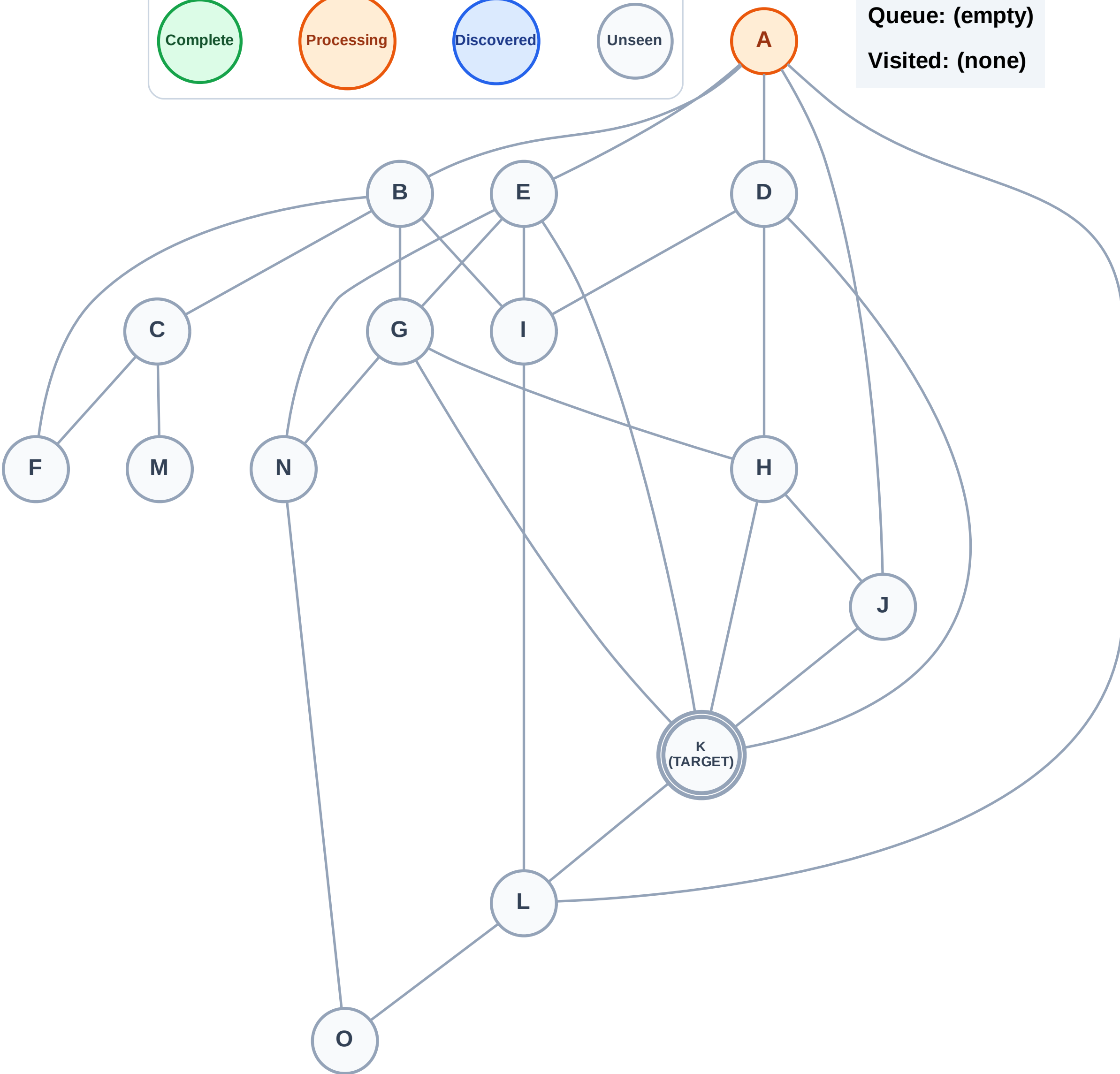
Discovered



Unseen

Queue: (empty)

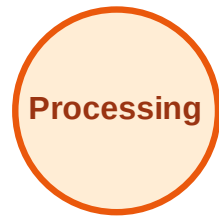
Visited: (none)



BFS Traversal

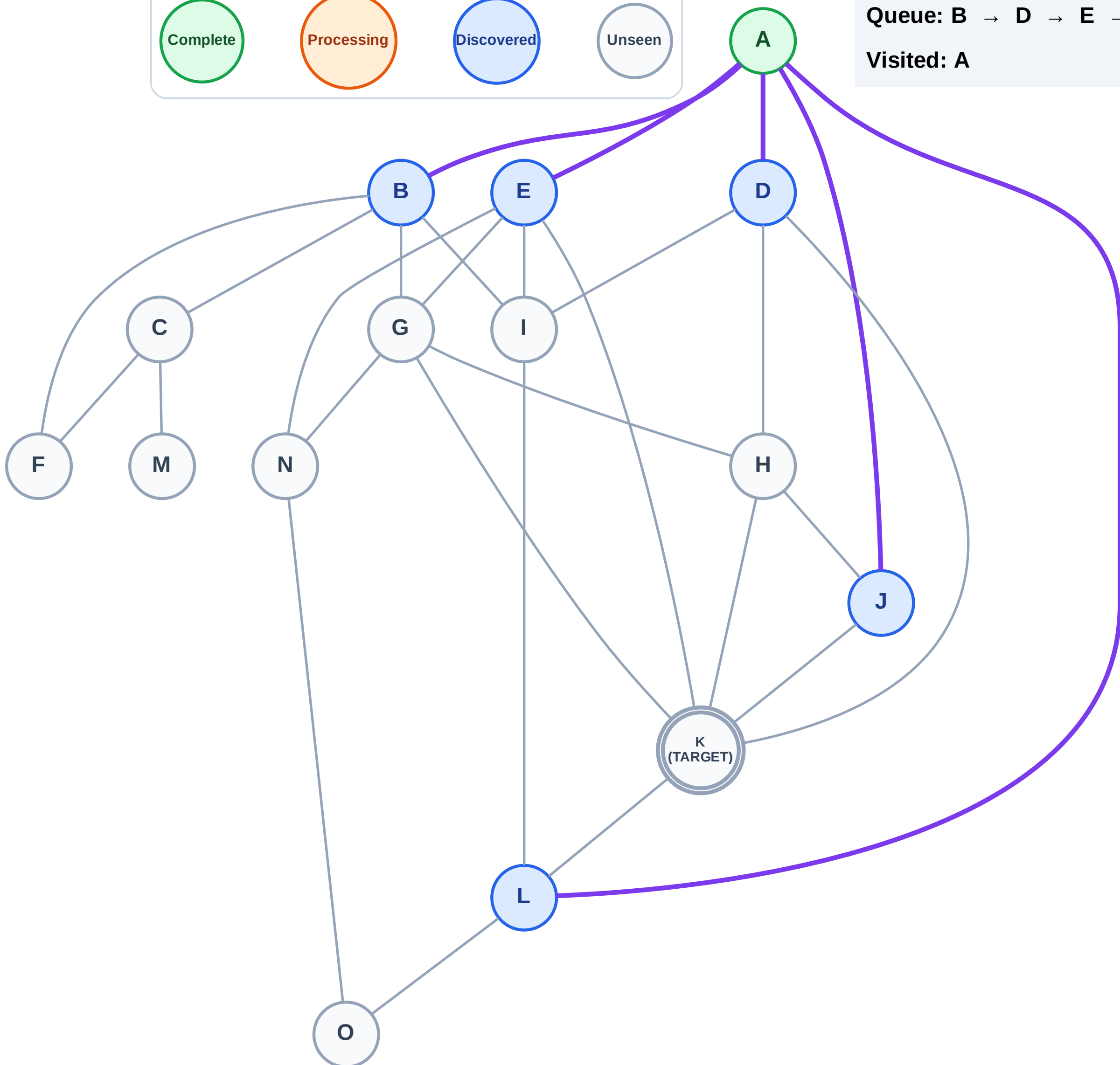
Step 3: Discover B, D, E, J, L from A

Legend



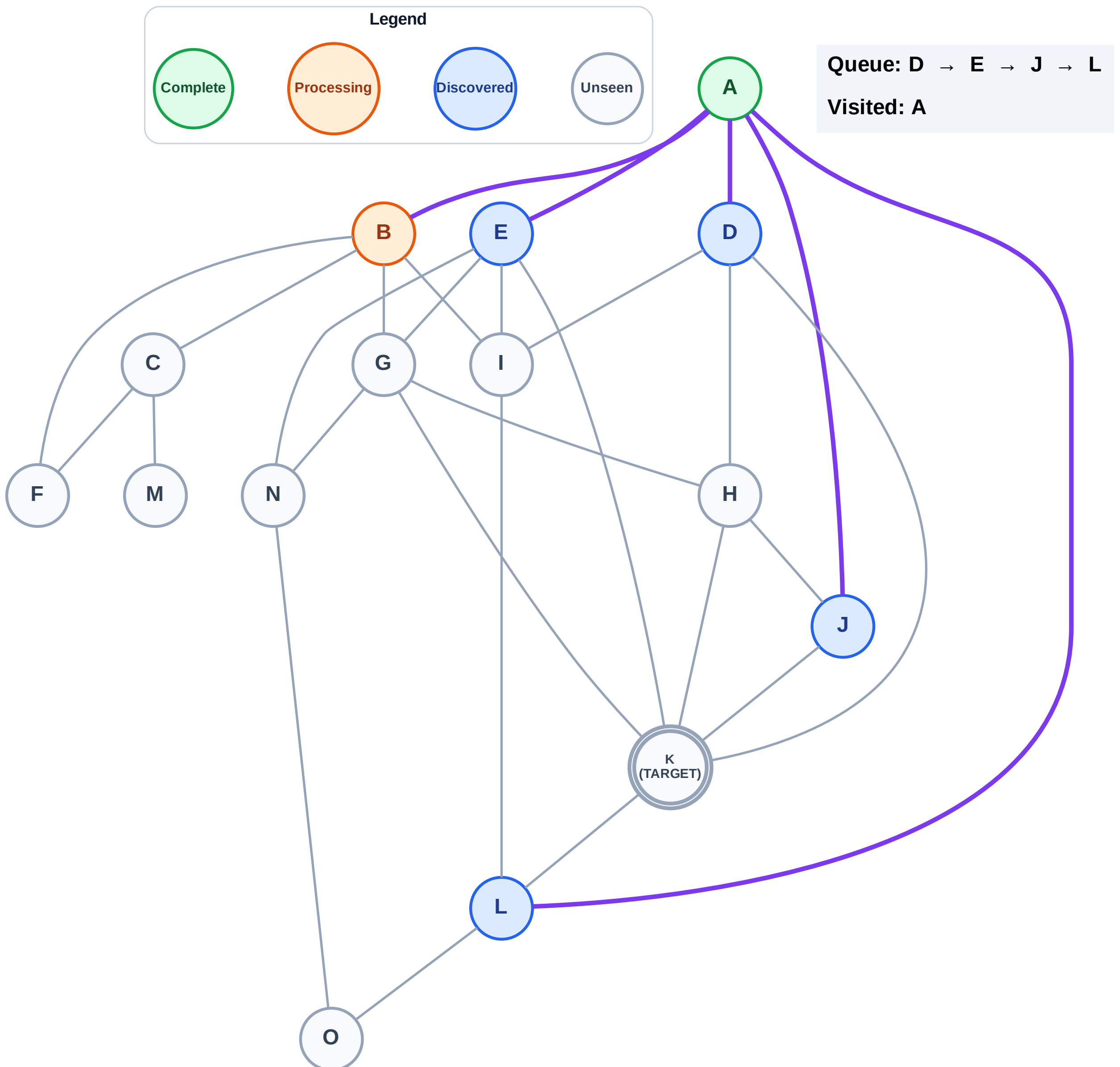
Queue: B → D → E → J → L

Visited: A



Step 4: Process node B

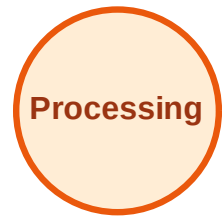
Step 4: Process node B



BFS Traversal

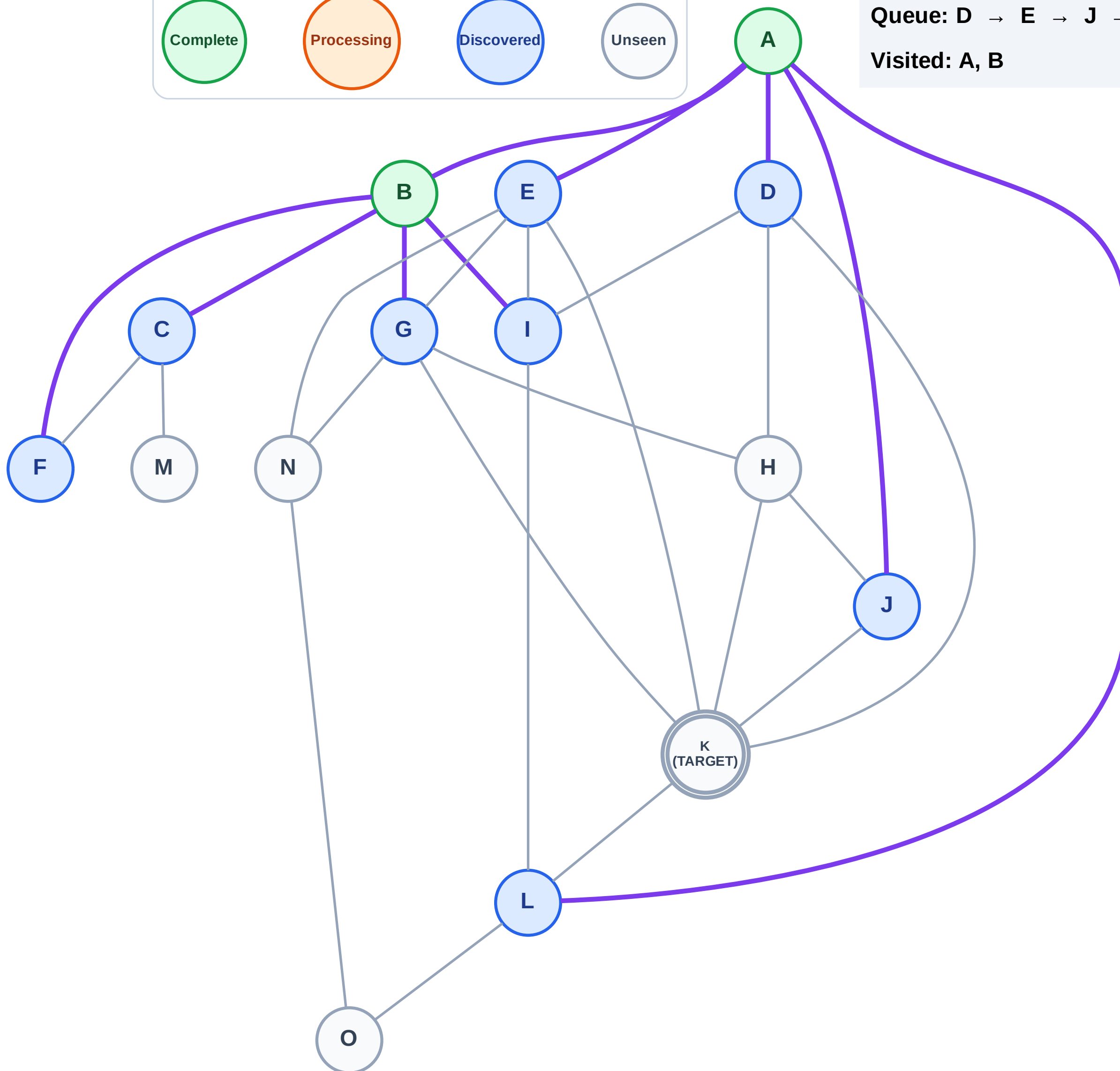
Step 5: Discover C, F, G, I from B

Legend



Queue: D → E → J → L → C → F → G → I

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

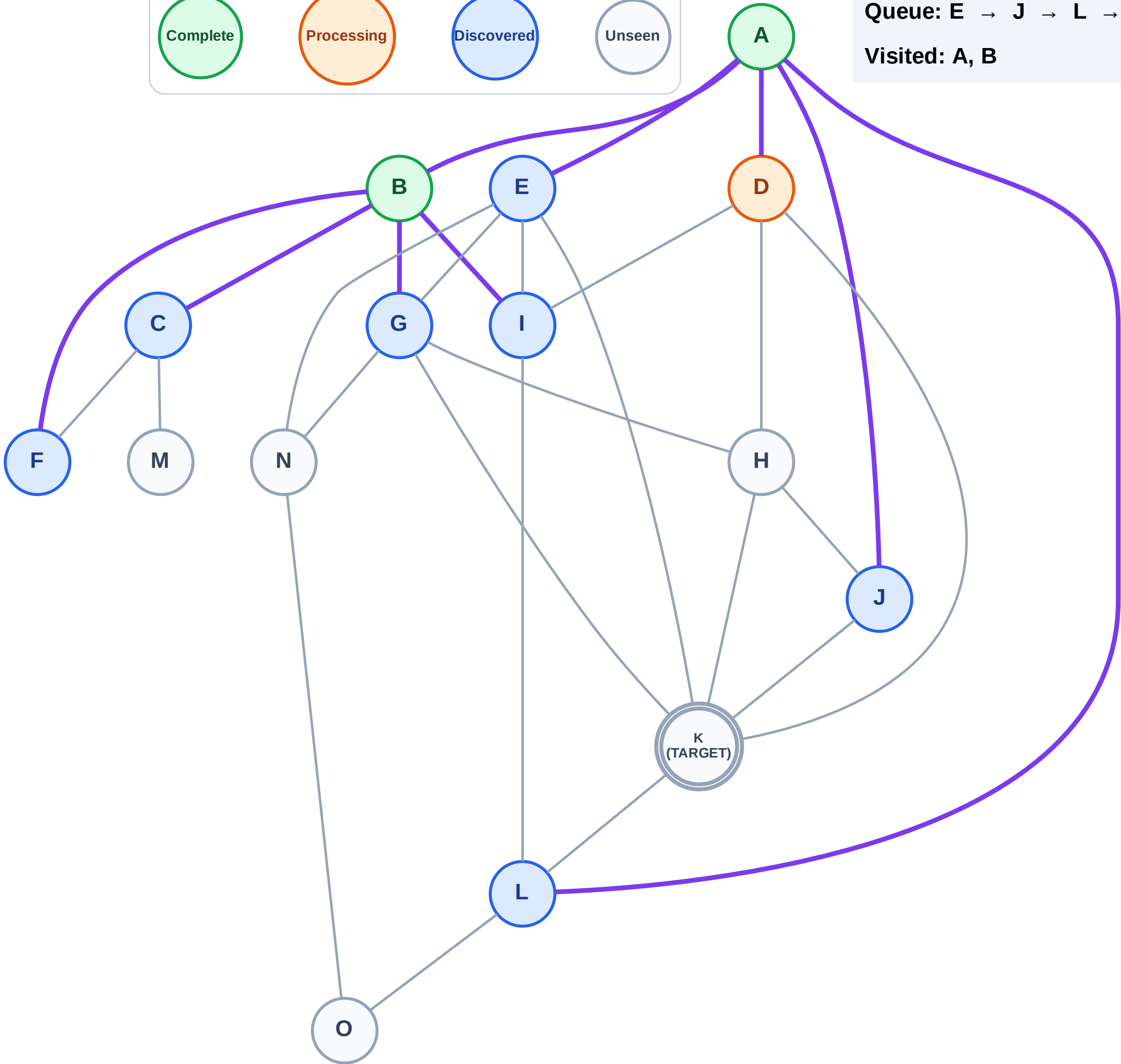
Processing

Discovered

Unseen

Queue: E → J → L → C → F → G → I

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend

Complete

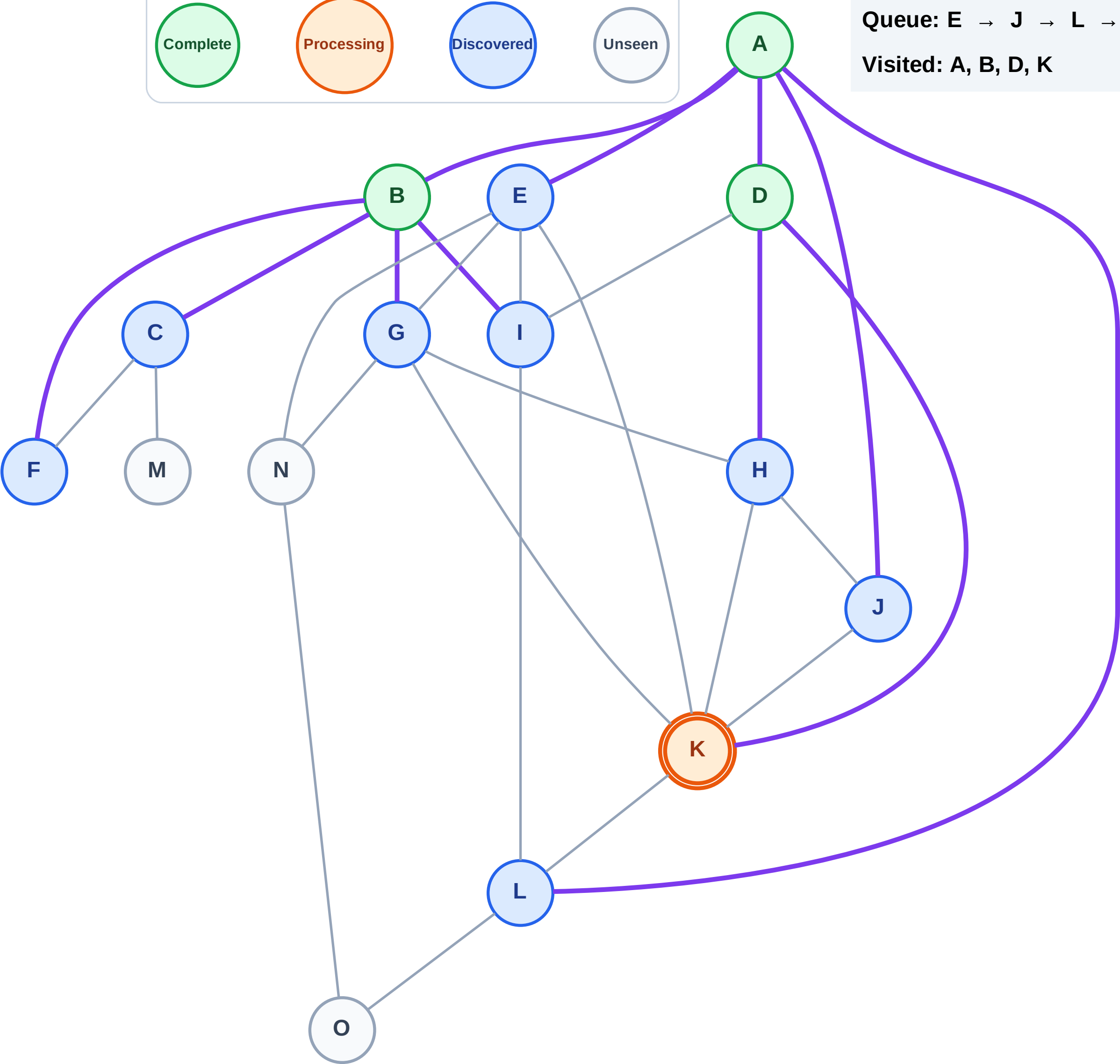
Processing

Discovered

Unseen

Queue: E → J → L → C → F → G → I → H

Visited: A, B, D, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

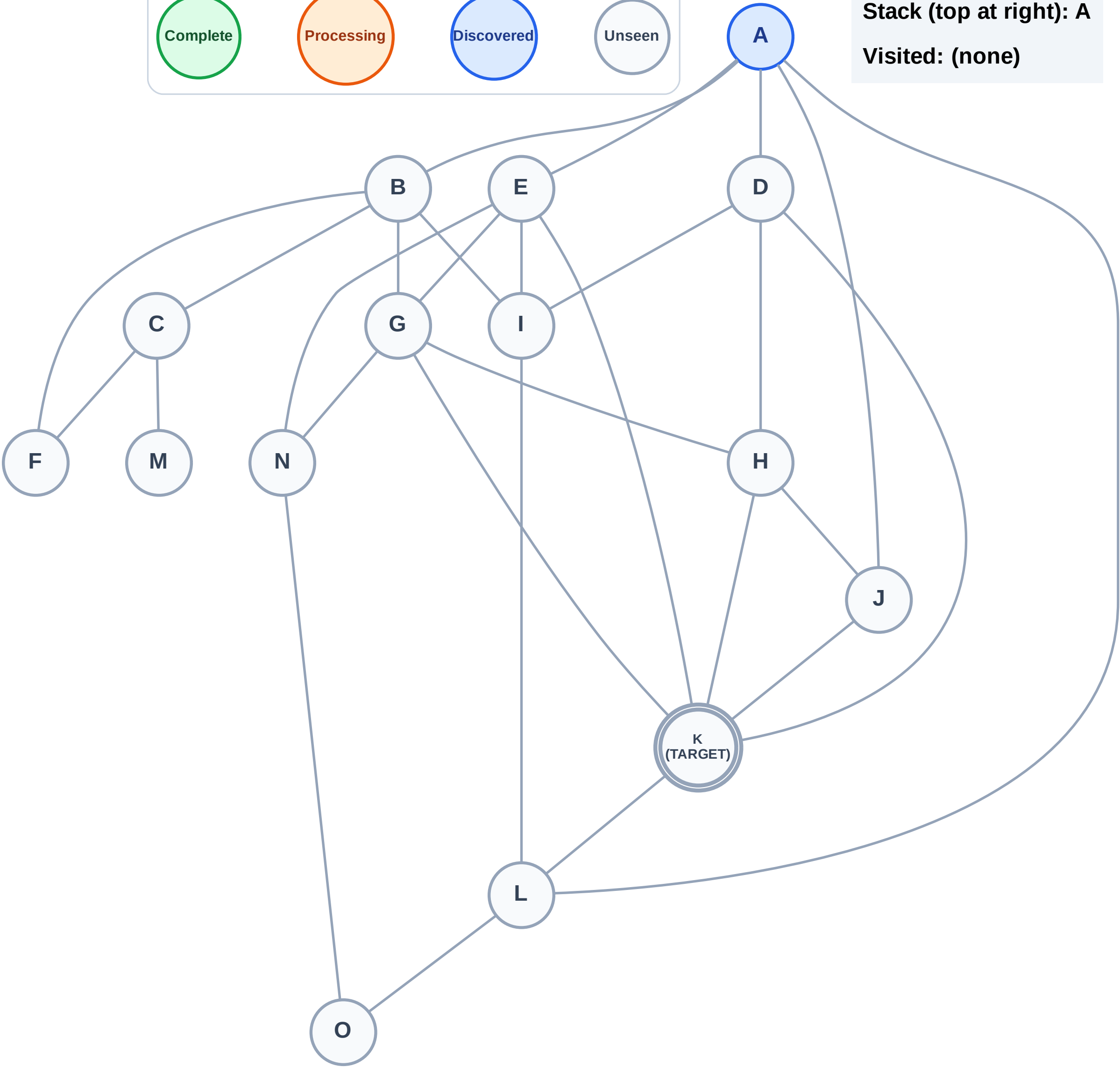
Complete

Processing

Discovered

Unseen

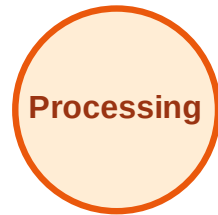
Stack (top at right): A
Visited: (none)



DFS Traversal

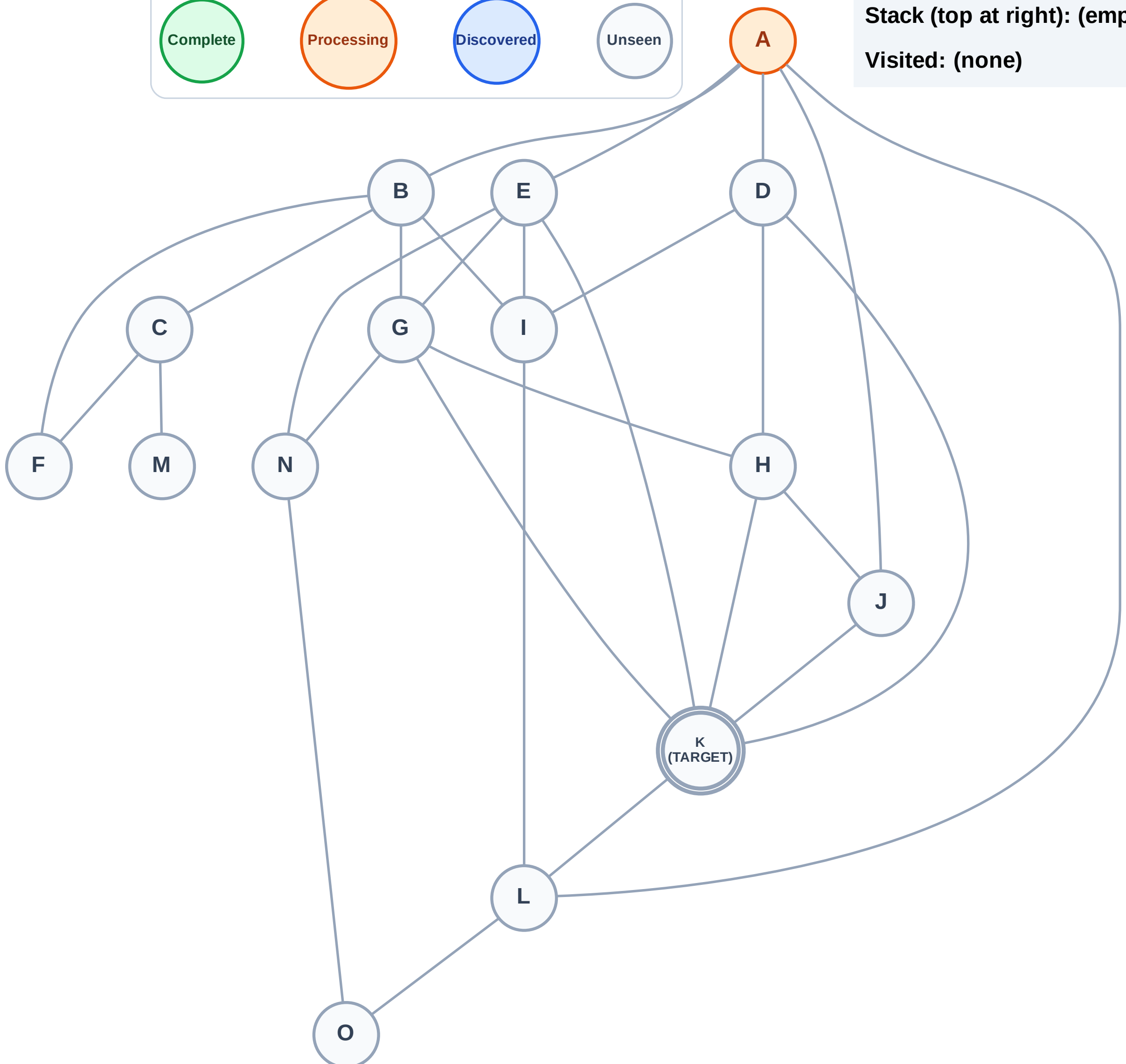
Step 2: Process node A

Legend



Stack (top at right): (empty)

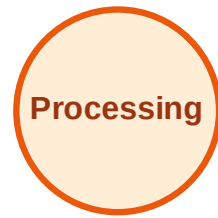
Visited: (none)



DFS Traversal

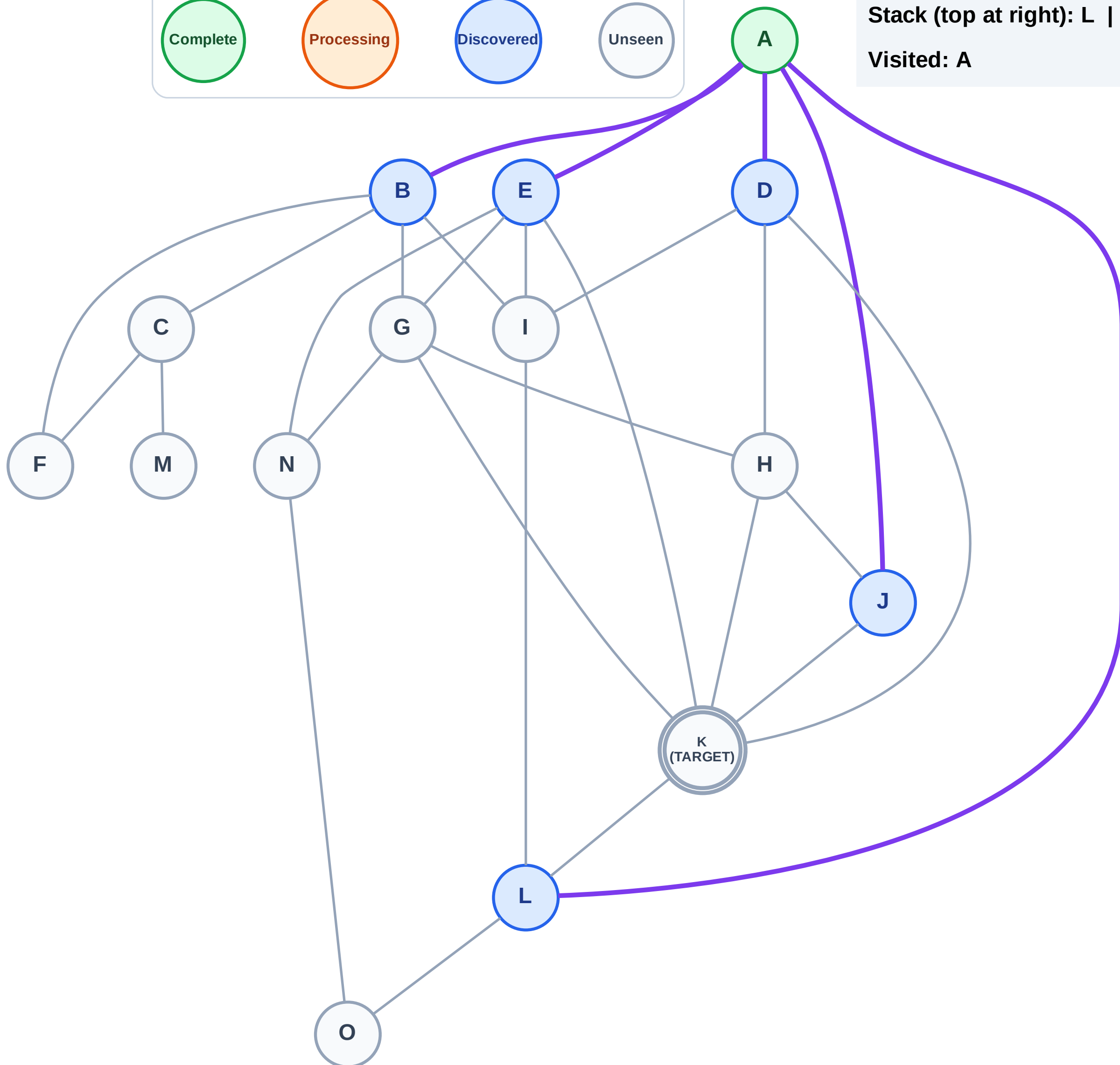
Step 3: Discover L, J, E, D, B from A

Legend



Stack (top at right): L | J | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

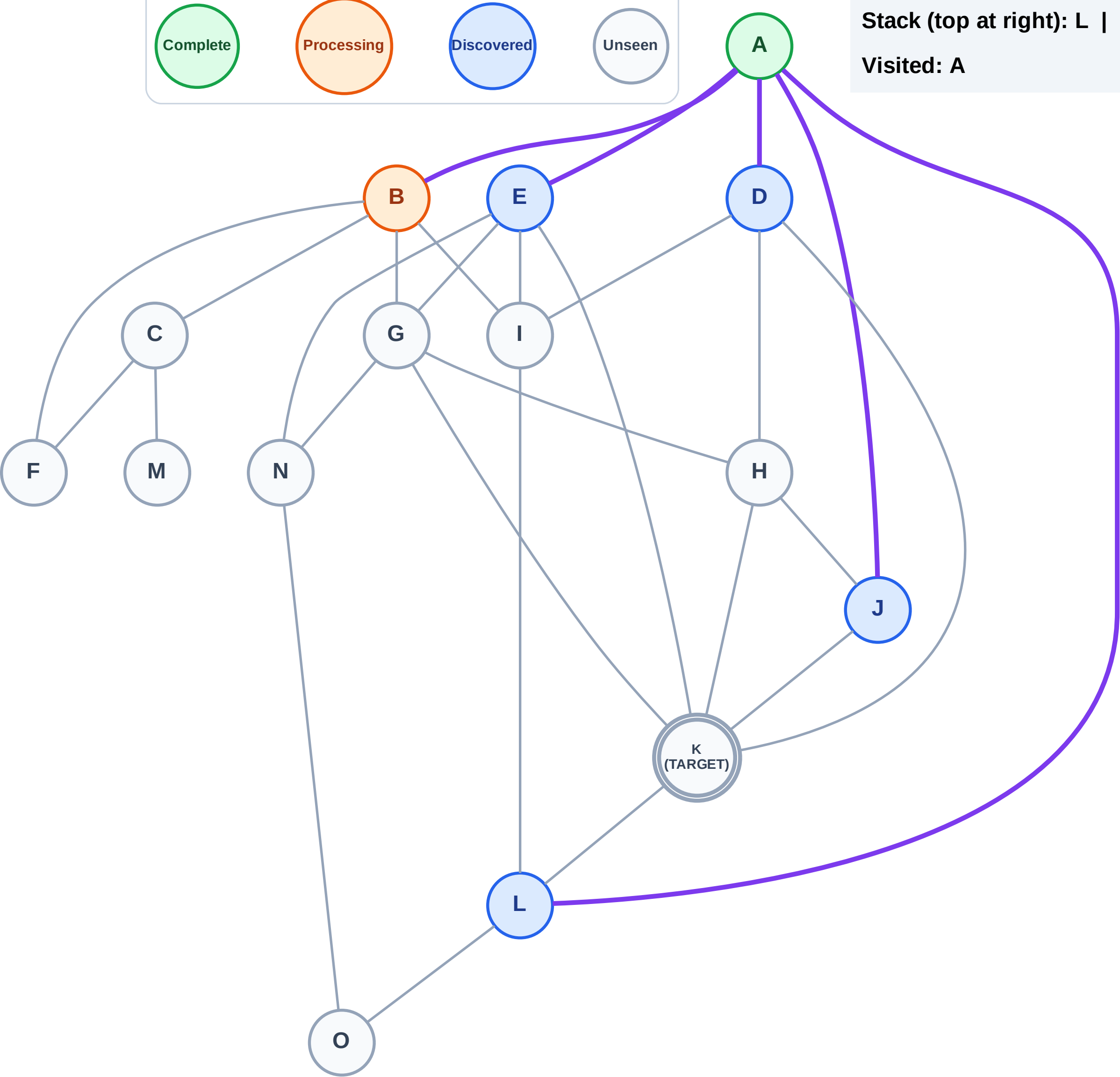
Processing

Discovered

Unseen

Stack (top at right): L | J | E | D

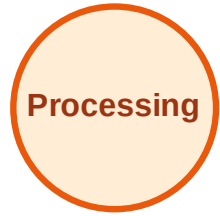
Visited: A



DFS Traversal

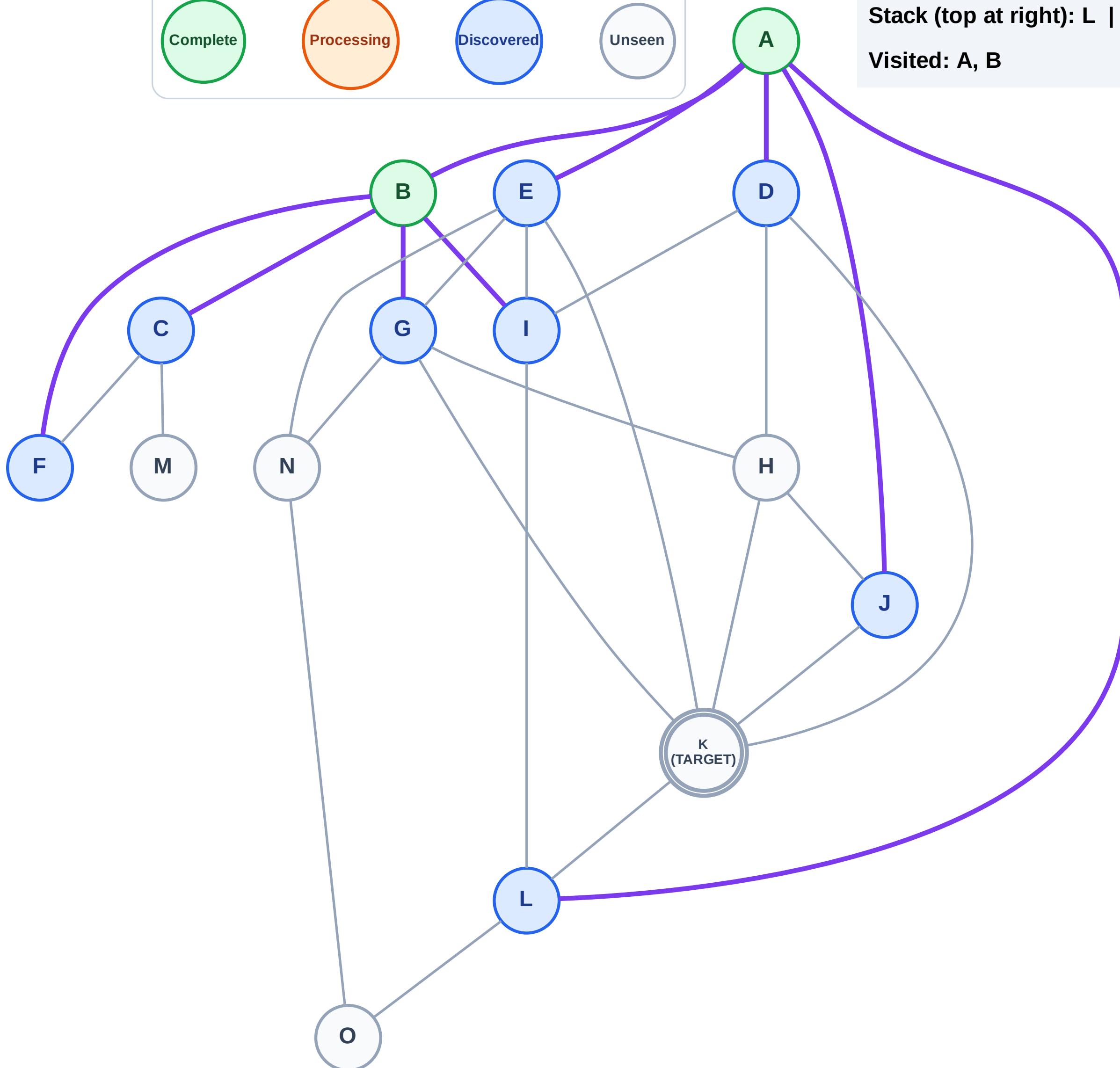
Step 5: Discover I, G, F, C from B

Legend



Stack (top at right): L | J | E | D | I | G | F | C

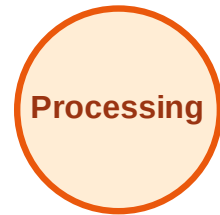
Visited: A, B



DFS Traversal

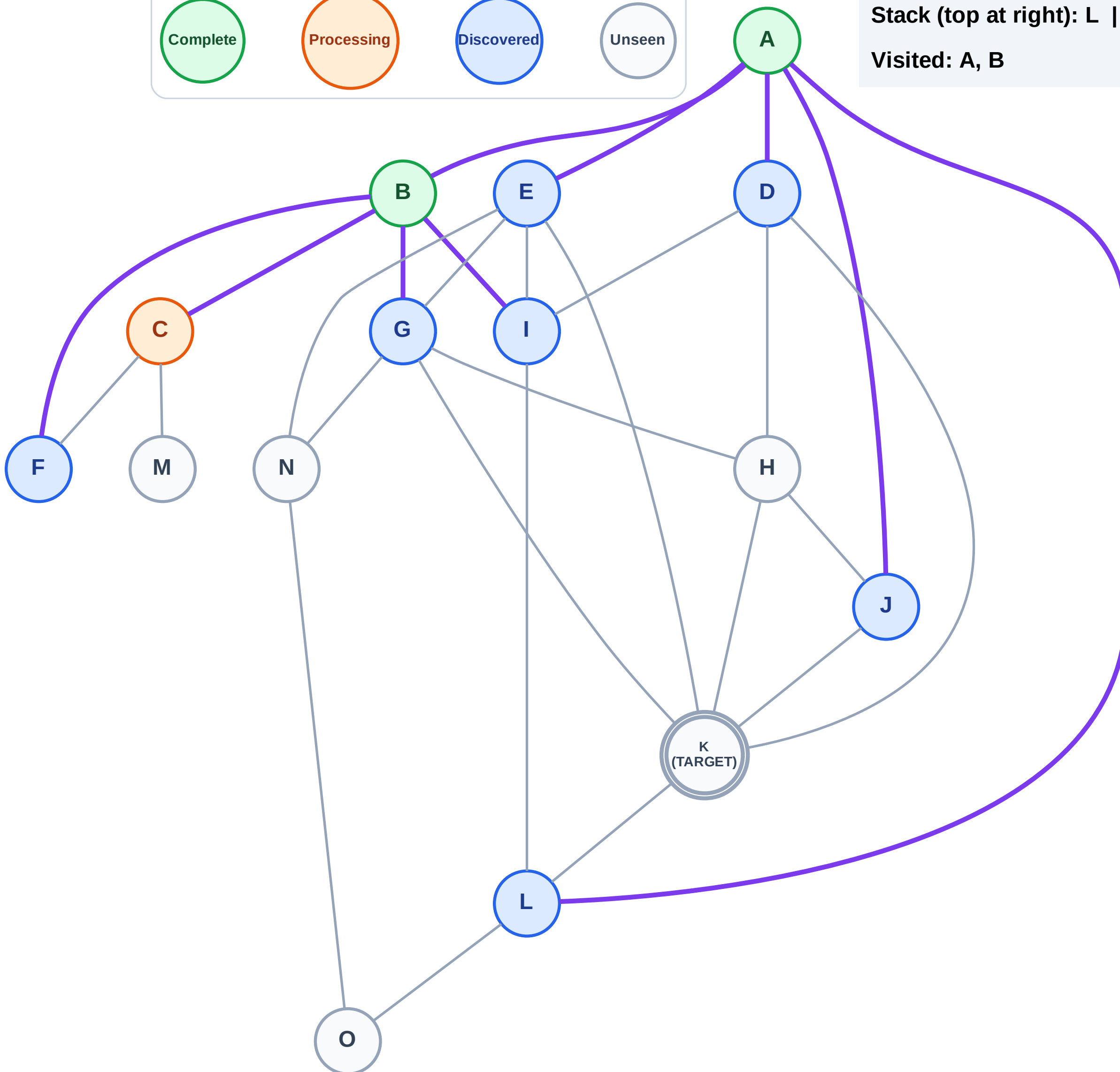
Step 6: Process node C

Legend



Stack (top at right): L | J | E | D | I | G | F

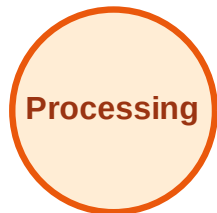
Visited: A, B



DFS Traversal

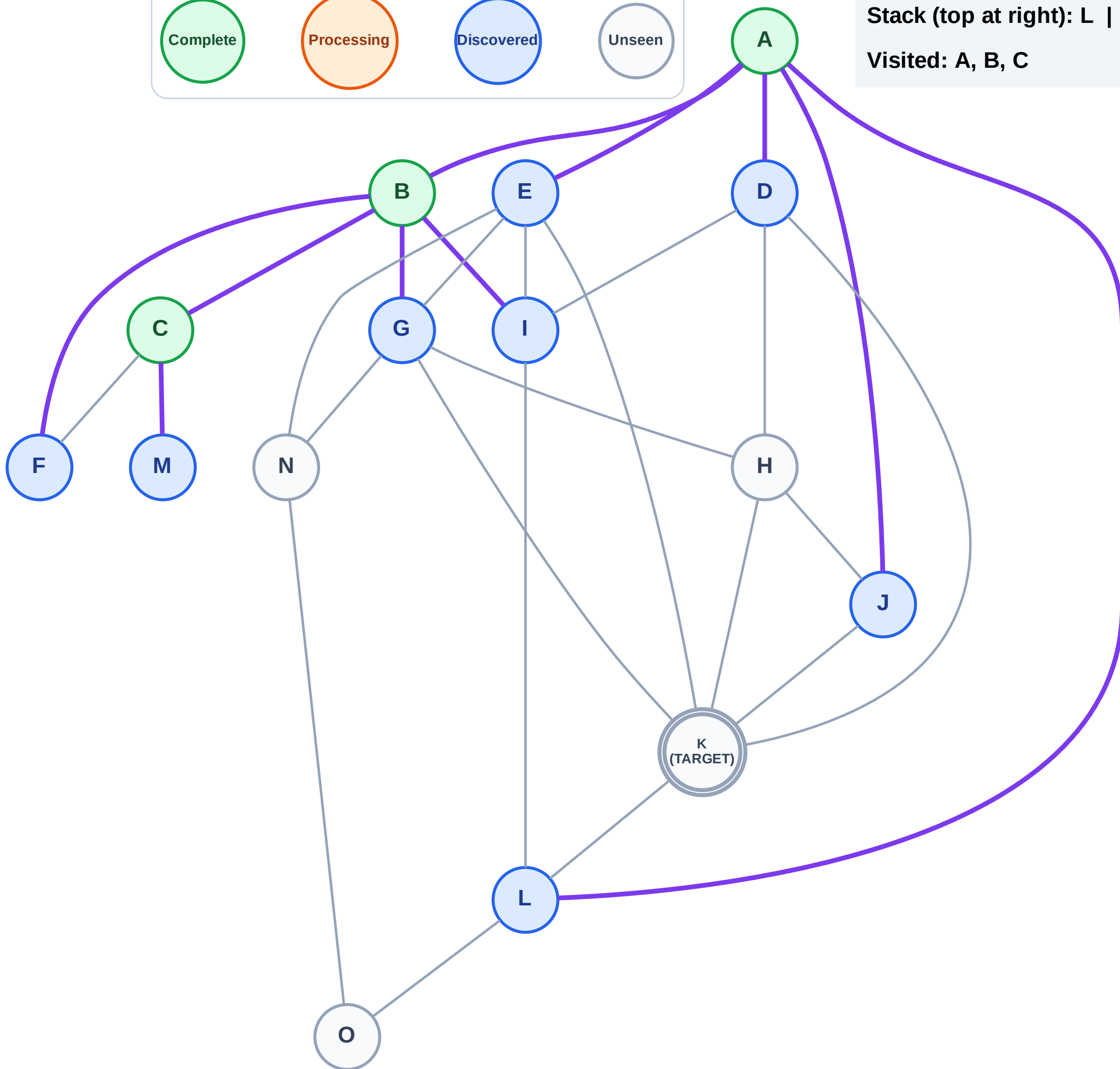
Step 7: Discover M from C

Legend



Stack (top at right): L | J | E | D | I | G | F | M

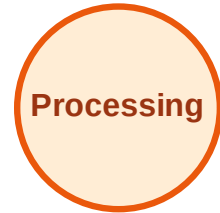
Visited: A, B, C



DFS Traversal

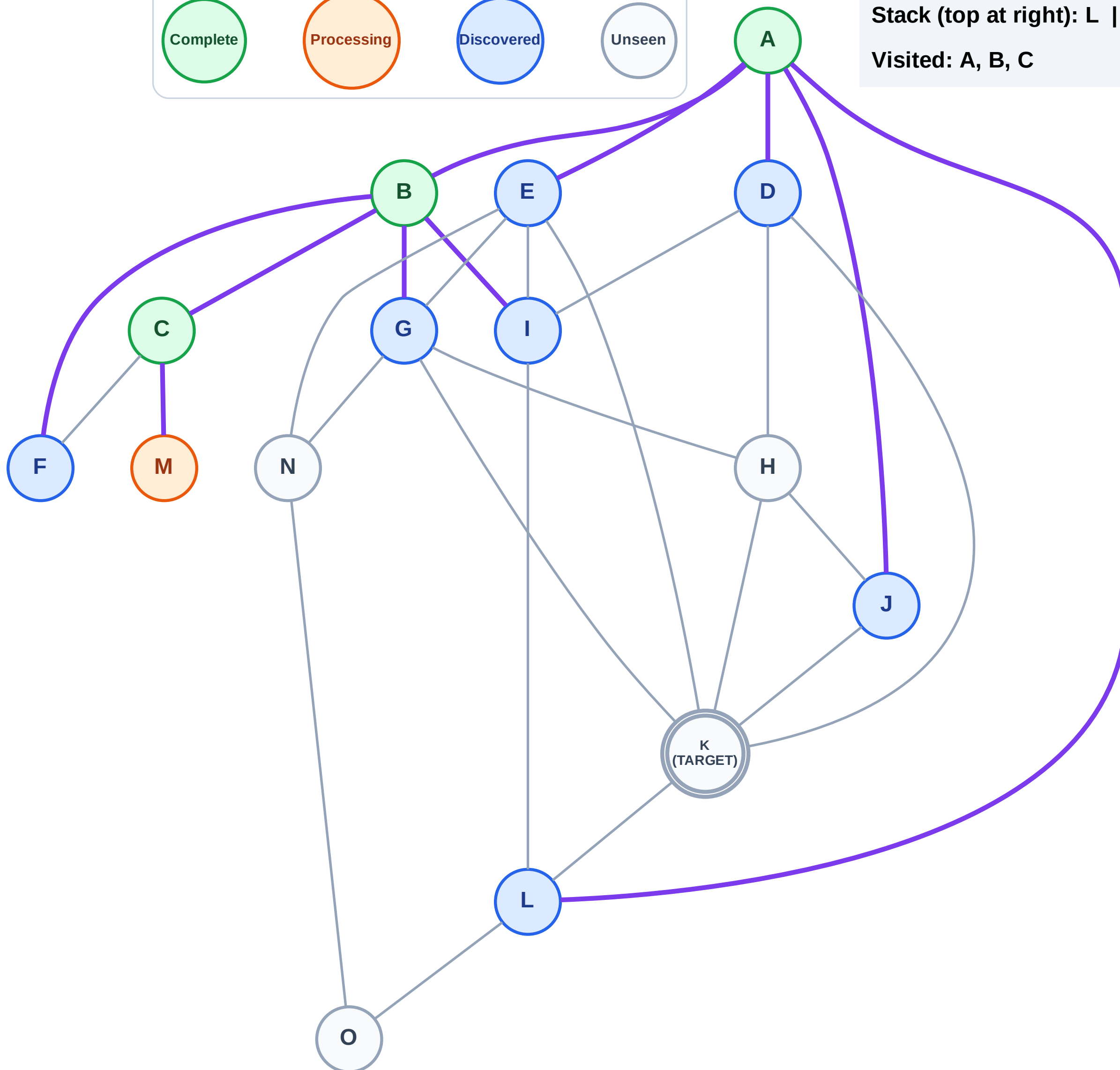
Step 8: Process node M

Legend



Stack (top at right): L | J | E | D | I | G | F

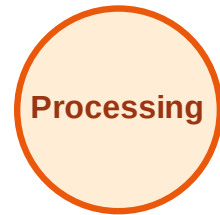
Visited: A, B, C



DFS Traversal

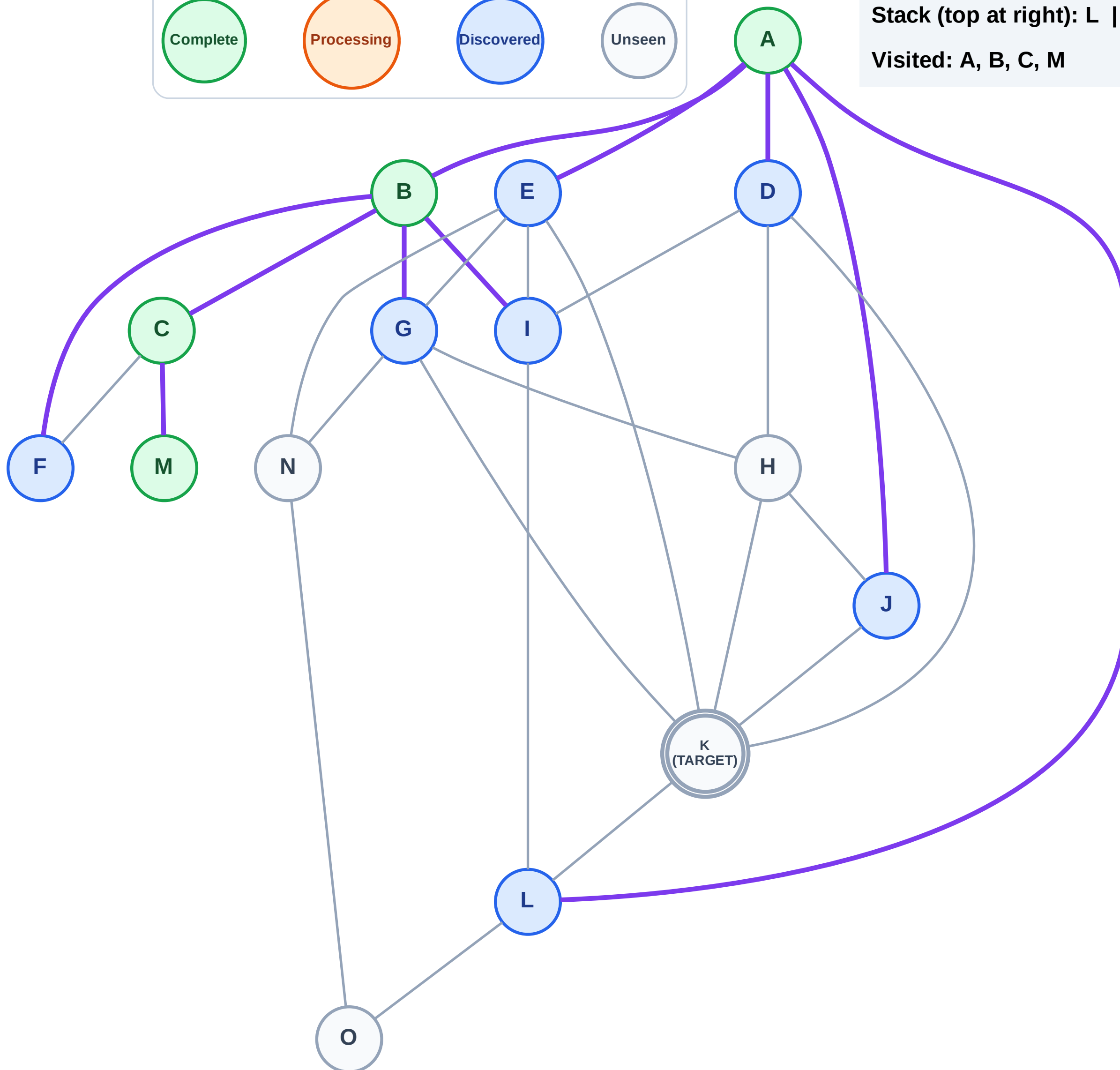
Step 9: No new neighbors from M

Legend



Stack (top at right): L | J | E | D | I | G | F

Visited: A, B, C, M



DFS Traversal

Step 10: Process node F

Legend

Complete

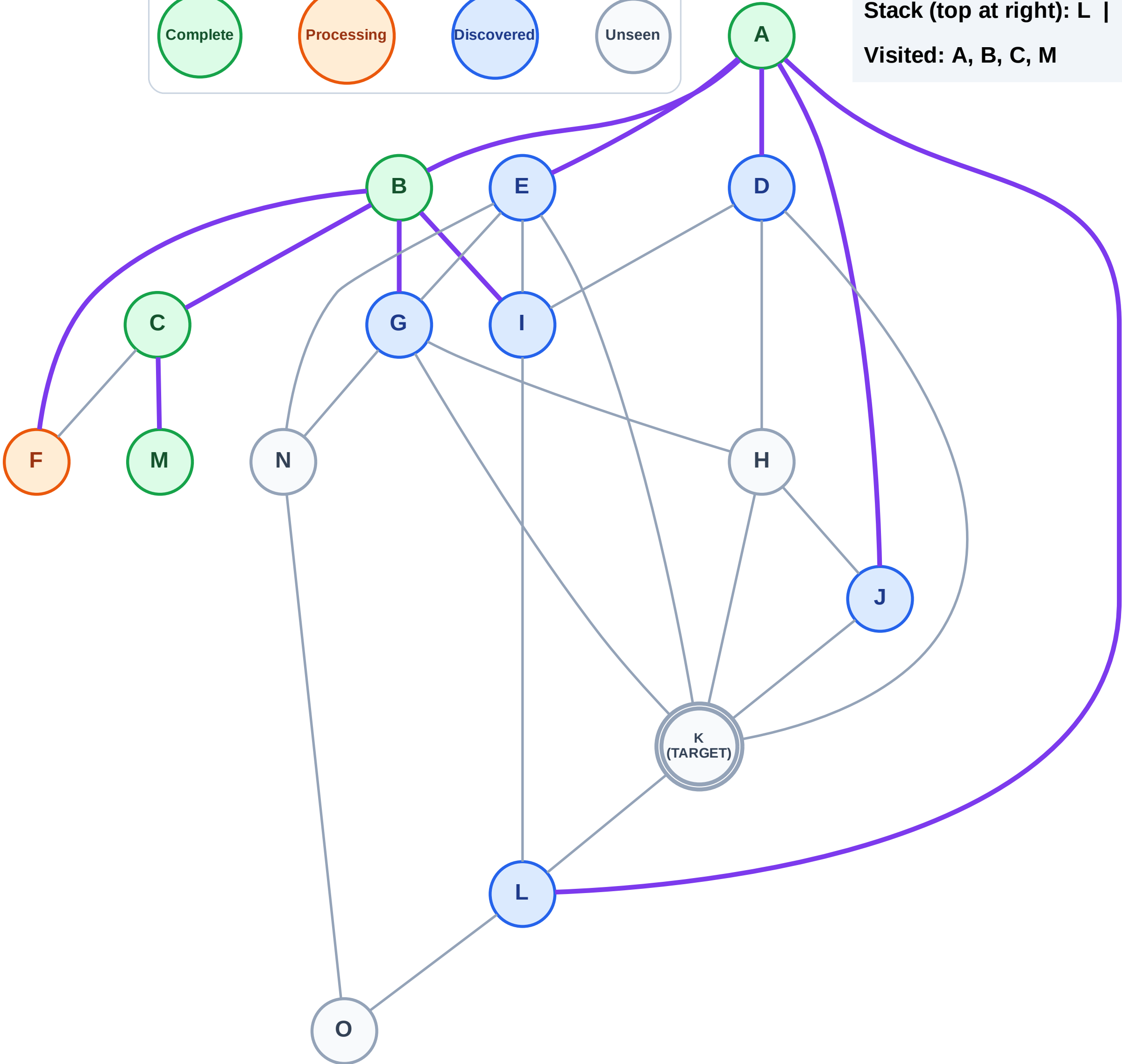
Processing

Discovered

Unseen

Stack (top at right): L | J | E | D | I | G

Visited: A, B, C, M



DFS Traversal

Step 11: No new neighbors from F

Legend

Complete

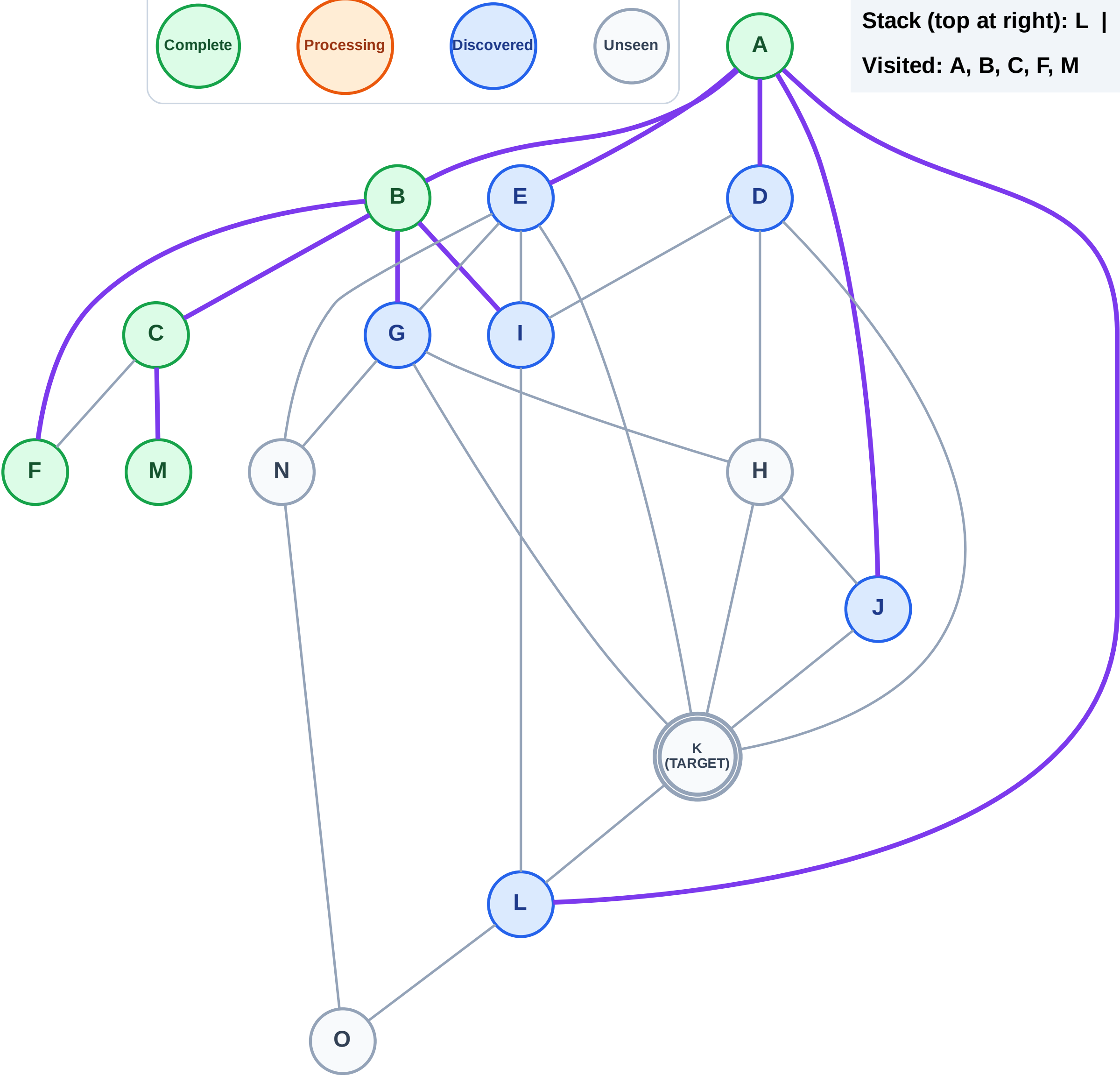
Processing

Discovered

Unseen

Stack (top at right): L | J | E | D | I | G

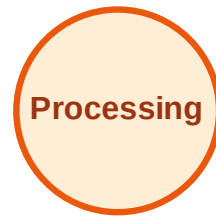
Visited: A, B, C, F, M



DFS Traversal

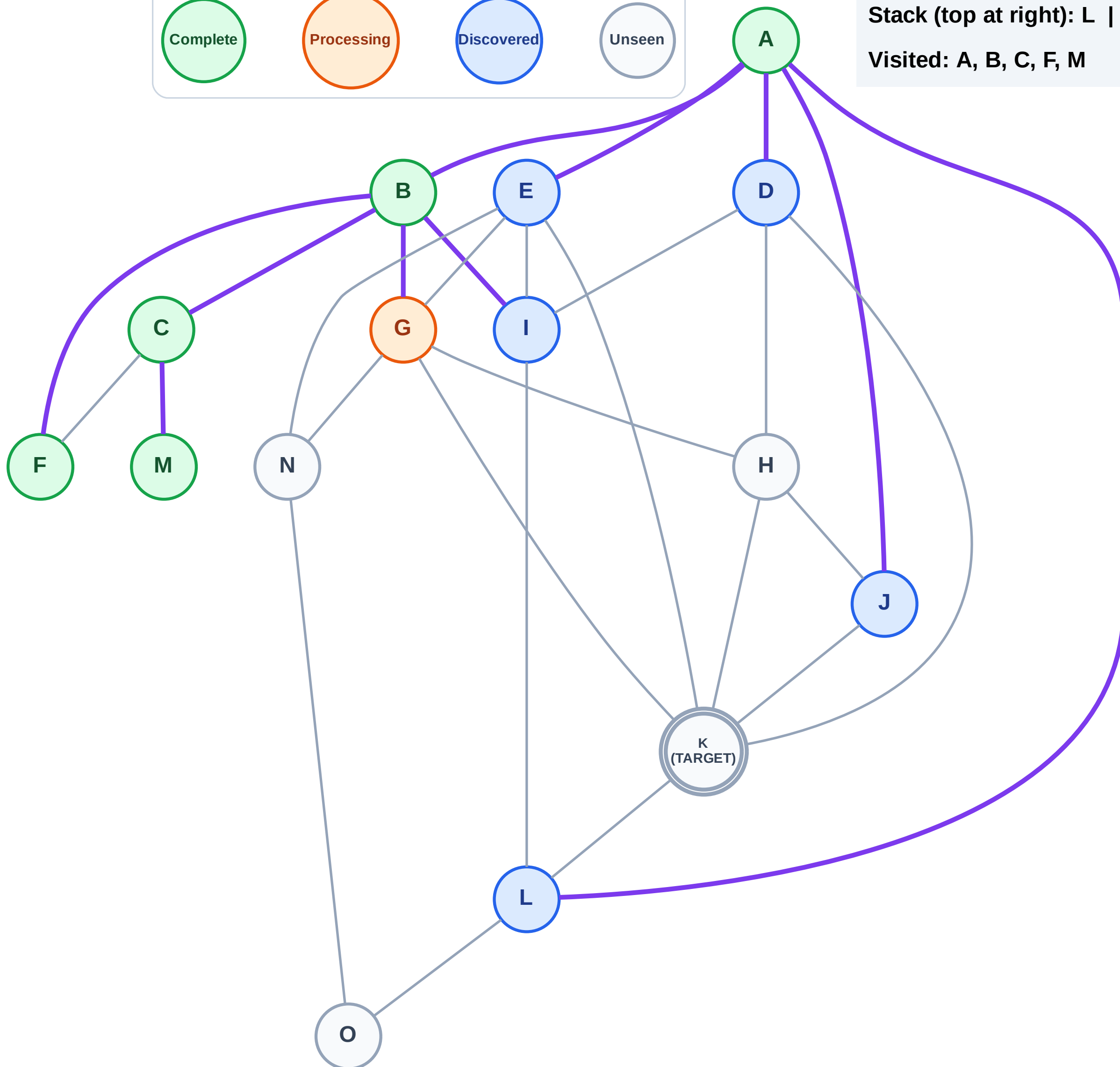
Step 12: Process node G

Legend



Stack (top at right): L | J | E | D | I

Visited: A, B, C, F, M



DFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | J | E | D | I | N

Visited: A, B, C, F, G, K, M

